

Course: B. Tech. Branch: Electrical Engineering / Electrical Engineering (Electronics and Power)/ Electrical & Electronics Engg. / Electrical & Power Engineering.

Subject Code & Name: (BTEEC601) Switchgear and Protection

Max Marks: 60

Date: 13 / 06 / 2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. Use of non-programmable scientific calculators is allowed.
3. Assume suitable data wherever necessary and mention it clearly.

Marks

Q. 1 Attempt any two (2) of the following.	12
A) Explain construction and working of balanced beam type relay with diagram.	6
B) Discuss primary and backup protection system	6
C) Explain HVDC circuit breaker.	6
Q.2 Attempt any two (2) of the following.	12
A) Explain the working principle of current differential relay	6
B) Explain impedance relay in detail	6
C) Discuss the following terms with neat waveform	6
i) Arc Voltage	
ii) Transient Restriking Voltage (TRV)	
iii) Rate of rise of Restriking Voltage (RRRV)	
iv) Recovery Voltage	
Q. 3 Attempt any two (2) of the following.	12
A) Explain arcing phenomenon circuit breakers.	6
B) Explain minimum oil circuit breaker.	6
C) Explain protection of ring main and parallel feeders system	6
Q.4 Attempt any two (2) of the following.	12
A) Explain operating principle of instantaneous over current relay	6
B) Explain carrier aided distance protection scheme.	6
C) What are advantages and disadvantages of static relays.	6

Q. 5	Attempt any two (2) of the following.	12
A)	Explain differential protection of alternator in detail.	6
B)	Explain percentage differential protection for Δ -Y transformer	6
C)	Explain SF6 circuit breaker.	6

***** End *****